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Telkom x RevoU
Virtual Internship Project

Customer Churn Prediction & Retention Strategy

From Prediction to Proactive Retention

Project 2 | Customer Analytics



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Business Context

Supporting sustainable growth through better customer retention

Customer retention is important for maintaining long-term business performance.

Currently, retention efforts are still mostly reactive, where customers are generally approached after they have submitted a cancellation request.

A more proactive approach can **identify customers who are likely to churn before they leave**.

Reactive Retention

Customers are generally approached after submitting a cancellation request.

Early Identification

Identify customers who are likely to churn before they leave.

Proactive Retention

Prioritize customers who may need further attention and take proactive actions.

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Business Problem

Identifying customers at risk before they churn

Current Challenge

The current retention process is still mostly reactive

Cancellation Request

Customers are generally identified for retention after submitting a cancellation request.

Retention Approach

The retention team approaches these customers only after the cancellation signal has occurred.

Late Identification

This makes it difficult to identify customers who are at risk of churning before they decide to leave.

Business Need

Identify customers with a higher likelihood of churn earlier, so retention efforts can be started before customers decide to leave.

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Business Objective & Business Questions

Defining the goal and key questions for churn prediction

Business Objective



To predict customers **at risk of churn** and provide **data-driven recommendations** to support proactive retention strategies.

Business Questions

1. Who should the business prioritize?
2. Why are these customers at risk?
3. Which customers should receive retention campaigns?
4. How can the model support proactive retention?

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Data & Analytical Approach

Using customer data and machine learning to identify churn risk

Dataset

Training Dataset

20,000

customers

Test Dataset

4,000

customers

Target: Churn

0 = Active | 1 = Churn

Analytical Approach

01 — Data Understanding
Understand customer data and assess data quality.

02 — Exploratory Analysis
Identify customer patterns related to churn.

03 — Data Preparation
Prepare data and engineer features for modeling.

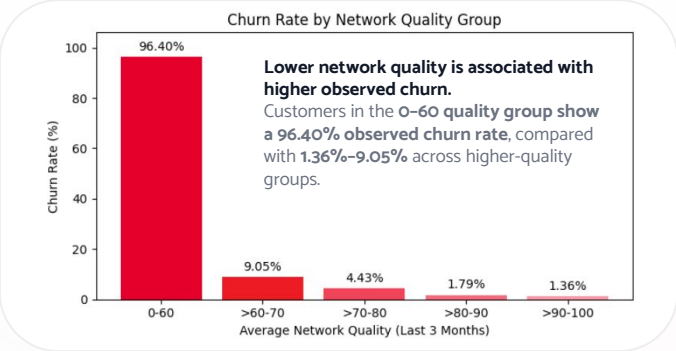
04 — Model Development
Compare and optimize classification models.

05 — Prediction & Explainability
Evaluate predictions and identify churn risk.

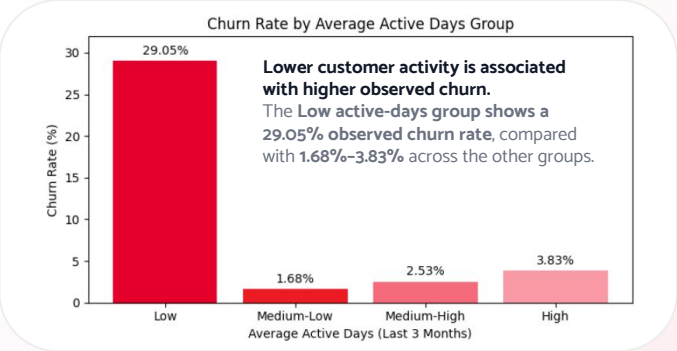
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Key Insights from Exploratory Data Analysis

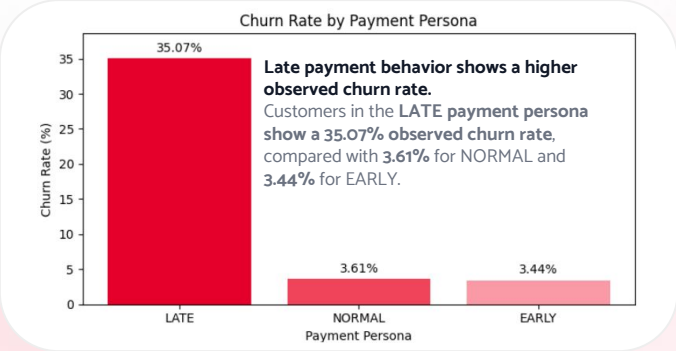
EDA reveals distinct patterns between churned and active customers



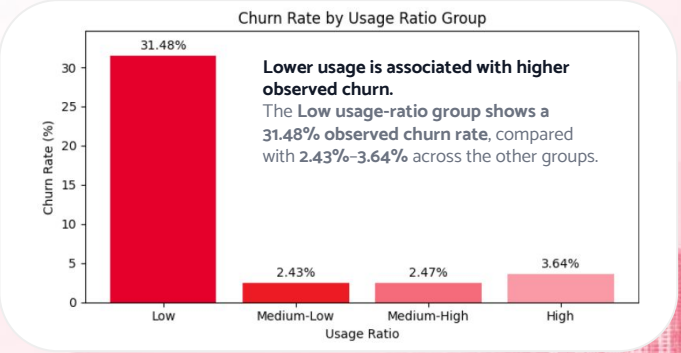
Lower network quality is associated with higher observed churn.
Customers in the **0–60 quality group show a 96.40% observed churn rate**, compared with **1.36%–9.05%** across higher-quality groups.



Lower customer activity is associated with higher observed churn.
The **Low active-days group shows a 29.05% observed churn rate**, compared with **1.68%–3.83%** across the other groups.



Late payment behavior shows a higher observed churn rate.
Customers in the **LATE payment persona show a 35.07% observed churn rate**, compared with **3.61%** for NORMAL and **3.44%** for EARLY.



Lower usage is associated with higher observed churn.
The **Low usage-ratio group shows a 31.48% observed churn rate**, compared with **2.43%–3.64%** across the other groups.

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Model Development & Performance

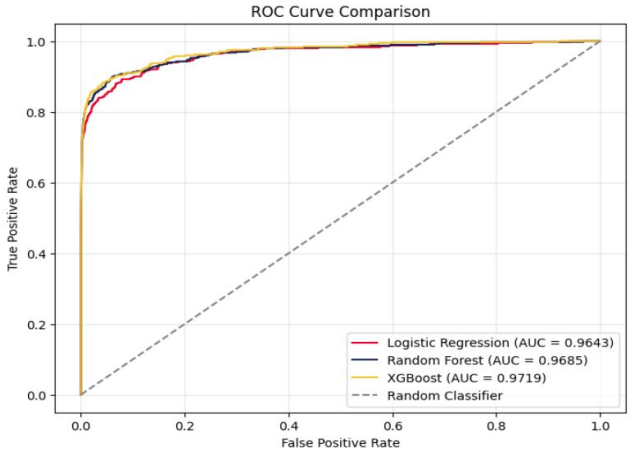
XGBoost was selected as the final model based on overall validation and cross-validation performance

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	96.90%	93.40%	74.25%	82.73%	96.43%
Random Forest	97.30%	93.98%	78.00 %	85.25%	96.85%
XGBoost	97.20%	92.60%	78.25%	84.82%	**97.19%**

XGBoost showed the highest **validation ROC-AUC (97.19%)** and **recall (78.25%)**, with a **CV ROC-AUC of 97.83%**.

ROC Curve (Test-Set)



XGBoost ROC-AUC: **97.19%**

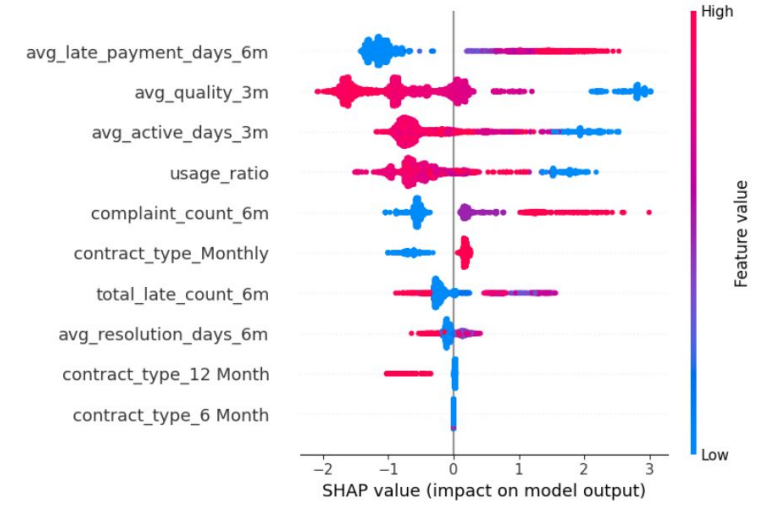
CV ROC-AUC: **97.83%**

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Key Drivers of Churn Prediction

SHAP analysis shows which features contribute most to the model's churn predictions

SHAP Summary Plot



SHAP values represent feature contribution to model predictions, not causal relationships.

Key Findings

01 **Network Quality | avg_quality_3m**
Most influential feature in the model's churn predictions.

02 **Payment Behavior | avg_late_payment_days_6m**
Higher values tend to push predictions toward Churn.

03 **Activity & Usage | avg_active_days_3m + usage_ratio**
Lower activity and usage values tend to increase churn prediction.

04 **Complaints | complaint_count_6m**
Higher complaint count tends to push predictions toward Churn.

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High-Risk Customers

The model identified 324 customers with a predicted churn probability of at least 50%

Prediction Results (Test Dataset)

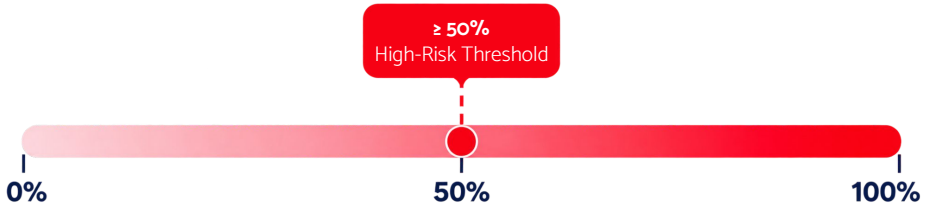
324
High-Risk Customers
8.10% of the test dataset

3,676
Predicted as Active
91.90% of the test dataset

Out of **4,000** test customers, **324** were predicted as high-risk for churn.

High-Risk Threshold

Customers with a predicted churn probability of at least 50% are classified as high-risk.



324 customers met the high-risk threshold and were prioritized for retention intervention.

Prediction Output

Each customer receives a predicted churn class and churn probability, enabling risk-based retention prioritization.

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Targeted Customer Groups

High-risk customers show different strongest SHAP drivers

Strongest SHAP Driver among High-Risk Customers

- **Network Quality** | avg_quality_3m | 287 | 88.58%
- **Payment Delays** | avg_late_payment_days_6m | 29 | 8.95%
- **Complaints** | complaint_count_6m | 3 | 0.93%
- **Late Payment Frequency** | total_late_count_6m | 3 | 0.93%
- **Active Days** | avg_active_days_3m | 2 | 0.62%

Key Takeaway

287

of 324 high-risk customers (88.58%)

Network quality is the dominant driver among high-risk customers.

SHAP Interpretation

All identified strongest drivers had **positive SHAP contributions**, meaning they pushed the model's predictions toward **Churn**.

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Business Recommendations

From churn prediction to targeted and proactive retention

01. Prioritize High-Risk Customers

Focus on **324 customers with ≥50% predicted churn probability**.

324

High-risk customers
with ≥50% predicted churn probability.

02. Address Key Risk Drivers

Tailor interventions based on each customer's **strongest SHAP driver**.

- **Network Quality** — Service / network recovery
- **Payment Delays** — Payment support and flexibility
- **Complaints** — Faster complaint resolution
- **Active Days** — Engagement and reactivation

03. Apply Targeted Retention Campaigns

Match retention actions to each customer's **risk profile and key driver**.

Target retention efforts based on each customer's strongest SHAP driver.

04. Operationalize Proactive Retention

Turn model predictions into a continuous retention process.



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Project Resources

Explore the project and my data analytics journey

Google Colab
Explore the complete Python workflow, from data analysis to churn prediction.

View Colab↗

Project Portfolio
Explore my data analytics portfolio and previous projects.

View Portfolio ↗

Presentation Recording
Watch the project presentation and analysis walkthrough.

Watch Recording ↗

LinkedIn
Connect with me and explore my data analytics journey.

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Thank you for your time and attention!

Eva Arliana
Aspiring Data Analyst | Turning data into actionable insights

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